

PRARTHANA KRISHNAMURTHY

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Summary

Machine learning engineer with 3+ years building production ML for high-stakes decisions across drug discovery, financial fraud, and generative text. Delivers calibrated, explainable, and monitored models end to end, from data pipelines and distributed training to deployed services, including fraud models scoring 500K transactions per day.

Experience

Machine Learning Engineer

Apr 2026 – Jul 2026

Alkermes Inc.

Waltham, MA

- Developed multi-task Chemprop and gradient-boosted ADMET models across 8 endpoints, improving prediction accuracy by 22% over existing models on novel scaffolds
- Rebuilt model validation with scaffold-based and temporal splits and conformal prediction intervals, correcting a 20% overstatement in reported model accuracy
- Created an uncertainty-aware compound prioritization model, reducing compounds needed to reach target potency by 30% across 2 drug discovery programs
- Deployed containerized model services across cloud and on-premise HPC with Docker, routing GPU and CPU workloads to separate pools for interactive and batch scoring

Machine Learning Researcher

Oct 2024 – Mar 2026

Northeastern University

Boston, MA

- Built a conditional text diffusion model generating full sentences from subject, verb, and object inputs in frozen BERT embedding space, decoding all positions in parallel through a 1D U-Net denoiser
- Engineered a pipeline extracting 400K training pairs from public-domain fiction with spaCy dependency parsing, filtering to single-clause sentences and handling passive voice and phrasal verbs
- Scaled training to 4 A100 GPUs on the university HPC cluster with PyTorch DDP and bf16 mixed precision, documenting 88% scaling efficiency across 1, 2, and 4 GPUs for the cluster's access review
- Restructured the codebase into config-driven modules with Weights & Biases tracking and DVC versioning, making every reported result reproducible from one config file
- Cut sampling from 1,000 to 20 network evaluations by replacing ancestral DDPM sampling with DDIM and second-order DPM-Solver++, staying within 1 BLEU point of the full-step baseline
- Benchmarked the diffusion model against a fine-tuned BART-base baseline, measuring 2.4x lower batch generation latency for 64-token outputs at 20 steps against a 3-point BLEU gap

Software Engineer, ML

Aug 2021 – Dec 2023

Societe Generale Global Solution Centre

Bangalore, India

- Developed fraud detection models for a card portfolio, from data exploration and feature engineering through real-time production scoring at 500K transactions per day
- Improved fraud detection precision by 18% at fixed recall by engineering transaction velocity features and NetworkX graph features over cardholder transaction networks
- Benchmarked LightGBM, XGBoost, and CatBoost for card fraud detection under purged walk-forward validation to stop delayed chargeback labels leaking, shipping LightGBM at 0.91 AUC and 0.58 AUPRC
- Integrated a SHAP explainability layer generating per-prediction feature attributions for compliance reporting, cutting manual audit time by 40% across flagged transaction reviews
- Built input and score drift monitoring with population stability index checks and automated retraining triggers, flagging model degradation three weeks before label-based metrics could
- Optimized the nightly feature pipeline from four hours to ninety minutes by repartitioning skewed joins and switching rolling aggregates to incremental computation

Software Engineer, ML Intern

Mar 2021 – May 2021

Capgemini

Bangalore, India

- Delivered a retail banking churn proof of concept spanning SQL analysis of 250K customers, calibrated modeling, and a stakeholder dashboard
- Fine-tuned a DistilBERT classifier routing banking customer-service emails across 12 intent categories, lifting macro-F1 from 0.78 with a TF-IDF and logistic regression baseline to 0.86

Publications

- [Non-Autoregressive Text Generation Using Diffusion Models](#), Master's Thesis, Northeastern University

Skills

Languages: Python, SQL, Java, Bash
Machine Learning and Generative AI: Hugging Face Transformers, Scikit-learn, Optuna, Reinforcement Learning, Transfer Learning, Non-Autoregressive Generation, MCP
MLOps, Data, and Serving: Kubernetes, Terraform, MLflow, Airflow, GitLab CI/CD, Prometheus, PySpark, PostgreSQL, FastAPI
Evaluation and Statistics: Champion-Challenger, Shadow Deployment, Active Learning, Bayesian Optimization, Anomaly Detection, Class Imbalance, Statistical Modeling, Hypothesis Testing
Drug Discovery and Cheminformatics: RDKit, REINVENT4, QSAR Modeling, Molecular Fingerprints (ECFP, MACCS), Virtual Screening, Message-Passing Neural Networks

Education

MS in Computer Software Engineering , Northeastern University, GPA: 3.84/4.0	Jan 2024 – Dec 2025
BE in Electronics and Communication , Visvesvaraya Technological University	Aug 2017 – Aug 2021